

Timeline check in

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```
# data wrangling and visualization
library(tidyverse)

# allows file paths to work across computers
library(here)

# formatting axis labels
library(scales)

# working with dates
library(lubridate)

# combining plots
library(patchwork)
```

```
# read in bird survey data
birds <- read.csv(here("data", "birds.csv"))

# read in NOAA weather data
weather <- read.csv(here("data", "NOAA_daily_summaries.csv"))
```

Ideas for analysis and background

Research Questions

- What is the effect of previous month's precipitation on total monthly waterfowl abundance at NCOS?
- How does total waterfowl abundance vary through time at NCOS?
- Are there seasonal patterns in waterfowl abundance and precipitation at NCOS?

Relevance

Relevance to NCOS: NCOS is a restored coastal wetland in Santa Barbara, CA. Understanding what drives waterfowl abundance at this site helps evaluate whether restoration efforts are creating suitable habitat conditions for migratory and resident waterbirds.

Relevance to other systems: Precipitation-abundance relationships in waterfowl are relevant to other California estuaries and coastal wetlands, where water availability similarly structures habitat quality. Findings from NCOS can inform management decisions at comparable restored wetland systems throughout the region.

Relevance to understanding the organisms and physical processes: Waterfowl are strongly tied to wetland hydrology; precipitation influences water depth, food availability, and habitat extent, all of which affect where and when birds concentrate (Kahara et al. 2021; Stenzel and Page 2011). Understanding how rainfall patterns shape waterfowl abundance contributes to broader knowledge of how climate variability affects waterbird communities.

Planned Analysis

We plan to explore the relationship between lagged monthly precipitation and total monthly waterfowl abundance using a scatter plot and linear regression. We will also compare seasonal totals of both variables to assess whether seasonal patterns align across water years 2020–2025.

Citations

- **Habitat quality and drought effects on breeding mallard and other waterfowl populations in California, USA** (Kahara et al. 2021): This report shows how drought impacts habitat quality which in turn impacts waterfowl population. This is relevant to our study because we are looking at how precipitation impacts waterfowl abundance. Precipitation is a factor for drought.
- **Forecasting waterfowl population dynamics under climate change – Does the spatial variation of density dependence and environmental effects matter?** (Zhao et al. 2016): This report found that due to climate change, Mallard density is predicted to decline. One of the climate variables they examine is precipitation. This is relevant to our study because we are looking at the effects of precipitation on waterfowl. Their modelling is similar to what we are retroactively looking at.
- **The effect of climate change on optimal wetlands and waterfowl management in Western Canada** (Withey and Kooten 2011): This paper is relevant to our study because they are focusing on how climate change variables such as precipitation and

temperature changes will affect wetland habitats. This is where waterbirds live and precipitation affects the quality of the habitat.

- **Trends in abundance of wintering waterbirds relative to rainfall patterns at a central California estuary, 1972-2015** (Stenzel and Page 2011): This paper finds that there is a relationship between waterbirds and rainfall. They find that there is typically a negative relationship between the two, and that the relationship is stronger when looking at rainfall in the current year than in previous years. This is relevant to our study because we are looking at the relationship between waterfowl abundance and precipitation as well.

Preliminary code for analyses

Cleaning Data

```
birds_clean <- birds |>
# convert observation date to date format
mutate(
  observation_date = ymd(observation_date),

  # create monthly grouping variable
  month = floor_date(observation_date,
                     unit = "month")
) |>

# filter to only include waterfowl observations
filter(
  e_bird_group == "Waterfowl",

  # remove repeat observations
  repeat_observation == "No",

  # keep observations within study period
  observation_date > ymd("2019-09-30"),
  observation_date < ymd("2025-10-01")
)
```

```
weather_clean <- weather |>

# convert date column to date format
```

```
mutate(
  DATE = ymd(DATE),

  # create monthly grouping variable
  month = floor_date(DATE,
                    unit = "month")
) |>

# filter to study period
filter(
  DATE > ymd("2019-09-30"),
  DATE < ymd("2025-10-01")
)
```

Summaries

```
# calculate total monthly precipitation
monthly_precip <- weather_clean |>

# group observations by month
group_by(month) |>

# sum precipitation within each month
summarize(
  monthly_prctp = sum(PRCP, na.rm = TRUE),

  # remove grouping after summarizing
  .groups = "drop"
)
```

```
# calculate monthly abundance by species
monthly_waterfowl <- birds_clean |>

# group by month and species
group_by(month, species) |>

# sum bird counts within each month
summarize(
  abundance = sum(count, na.rm = TRUE),
```

```

# remove grouping after summarizing
.groups = "drop"
) |>

# order species factor levels for stacked area plot
# arranging so similar colors are not adjacent in stack
mutate(species = factor(species, levels = c(
  "Snow Goose",
  "Ross's Goose",
  "Cackling Goose (Aleutian)",
  "Canada Goose",
  "Greater White-fronted Goose",
  "Ring-necked Duck",
  "Redhead",
  "Canvasback",
  "Ruddy Duck",
  "American Wigeon",
  "Blue-winged Teal",
  "Cinnamon Teal",
  "Northern Pintail",
  "Mallard",
  "Northern Shoveler",
  "Bufflehead",
  "Hooded Merganser",
  "Green-winged Teal",
  "Gadwall"
)))

```

```

# calculate total monthly waterfowl abundance
total_monthly_waterfowl <- birds_clean |>

# group by month
group_by(month) |>

# sum all bird counts within each month
summarize(
  total_abundance = sum(count, na.rm = TRUE),

  # remove grouping after summarizing
  .groups = "drop"
)

```

Joining data

```
# join monthly bird abundance with monthly precipitation
bird_weather_monthly <- total_monthly_waterfowl |>

# join by month
left_join(monthly_precip,
          by = "month") |>

# arrange chronologically
arrange(month) |>

# create lagged precipitation and month-of-year variables
mutate(

  # previous month's precipitation (for figure 3)
  lagged_prctp = lag(monthly_prctp),

  # numeric month of year 1-12 (for figure 3 color scale)
  month_of_year = month(month),

  # season label derived from calendar month
  # (Dec/Jan/Feb = Winter, Mar/Apr/May = Spring,
  #   Jun/Jul/Aug = Summer, Sep/Oct/Nov = Fall)
  season = case_when(
    month(month) %in% c(12, 1, 2) ~ "Winter",
    month(month) %in% c(3, 4, 5) ~ "Spring",
    month(month) %in% c(6, 7, 8) ~ "Summer",
    TRUE ~ "Fall"
  ),
  season = factor(season, levels = c("Winter", "Spring", "Summer", "Fall"))
)

# step 1: sum abundance and precipitation within each season-year combination
seasonal_by_year <- bird_weather_monthly |>

# extract water year (Oct = start of new water year)
mutate(
  water_year = if_else(month(month) >= 10,
                      year(month) + 1,
                      year(month))
) |>
```

```

group_by(season, water_year) |>
summarize(
  # total waterfowl abundance for this season in this year
  seasonal_abundance = sum(total_abundance, na.rm = TRUE),

  # total precipitation for this season in this year
  seasonal_prctp = sum(monthly_prctp, na.rm = TRUE),

  .groups = "drop"
)

# step 2: average each season's totals across all water years
seasonal_summary <- seasonal_by_year |>

group_by(season) |>
summarize(
  avg_abundance = mean(seasonal_abundance, na.rm = TRUE),
  avg_prctp      = mean(seasonal_prctp, na.rm = TRUE),
  .groups = "drop"
)

```

Statistical analysis code

```

# linear regression: does previous month's precipitation
# predict total monthly waterfowl abundance?
abundance_model <- lm(total_abundance ~ lagged_prctp,
                      data = bird_weather_monthly)

summary(abundance_model)

```

Call:

```
lm(formula = total_abundance ~ lagged_prctp, data = bird_weather_monthly)
```

Residuals:

Min	1Q	Median	3Q	Max
-70.76	-40.71	-18.70	10.55	206.55

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	67.589	9.375	7.209	8.62e-10 ***
lagged_prdp	5.612	3.207	1.750	0.085 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 64.38 on 63 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared: 0.04635, Adjusted R-squared: 0.03121
F-statistic: 3.062 on 1 and 63 DF, p-value: 0.08501

The linear regression of total monthly waterfowl abundance on previous month's precipitation showed no significant relationship ($p > 0.05$), supporting the visual interpretation of Figure 3.

Visualizations directly relevant to answering questions

Figure 1: Monthly precipitation through time

```
ggplot(data = monthly_precip,
       mapping = aes(x = month,
                     y = monthly_prdp)) +

  # shaded area beneath line for depth
  geom_area(fill = "#4682B4", alpha = 0.25) +
  geom_line(color = "#7EC8E3", linewidth = 0.8) +

  scale_x_date(
    date_breaks = "1 year",
    date_labels = "%Y",
    expand = c(0, 0)
  ) +

  scale_y_continuous(
    expand = c(0, 0)
  ) +

  labs(
    title = "Monthly Precipitation Through Time",
    x      = "Date",
```



```

y      = "Monthly Precipitation (in)"
) +

theme_minimal(base_size = 13) +
theme(
  plot.background = element_rect(fill = "#1C2B2D", color = NA),
  panel.background = element_rect(fill = "#243336", color = NA),
  panel.grid.major = element_line(color = "#3A4F52", linewidth = 0.4),
  panel.grid.minor = element_blank(),
  plot.title = element_text(color = "#F0EAD6", size = 20, face = "bold",
                             margin = margin(b = 12), hjust = 0),
  axis.text = element_text(color = "#B8C9C2", size = 10),
  axis.title = element_text(color = "#D4C9A8", size = 11),
  axis.ticks = element_line(color = "#3A4F52"),
  plot.margin = margin(16, 20, 12, 16)
)

```

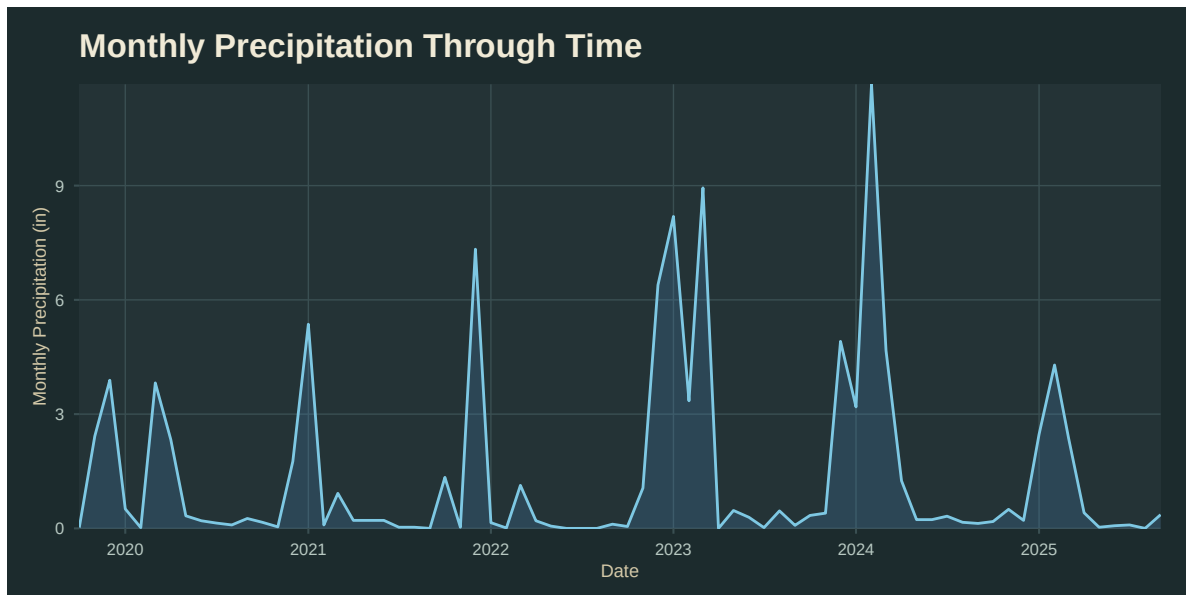


Figure 1 depicts total monthly precipitation (inches) at NCOS from water years 2020–2025, where the shaded area and line represent total precipitation summed across all days in each month. Precipitation follows a strongly seasonal pattern, peaking in winter months and dropping to near zero in summer. This figure provides environmental context for interpreting waterfowl abundance patterns, as precipitation may influence habitat conditions at NCOS.

Figure 2: Monthly waterfowl abundance through time by species

```
# color palette for all 19 species
waterfowl_colors <- c(
  "American Wigeon"           = "#C1440E",
  "Blue-winged Teal"         = "#1B6CA8",
  "Bufflehead"               = "#E8A838",
  "Cackling Goose (Aleutian)" = "#9DBF9E",
  "Canada Goose"             = "#6B8E23",
  "Canvasback"               = "#B22222",
  "Cinnamon Teal"            = "#A0522D",
  "Gadwall"                  = "#8FBC8F",
  "Greater White-fronted Goose" = "#D2B48C",
  "Green-winged Teal"        = "#2E8B57",
  "Hooded Merganser"         = "#20B2AA",
  "Mallard"                  = "#4682B4",
  "Northern Pintail"          = "#5F9EA0",
  "Northern Shoveler"         = "#3CB371",
  "Redhead"                  = "#8B1A1A",
  "Ring-necked Duck"         = "#6A5ACD",
  "Ross's Goose"             = "#F4E4B8",
  "Ruddy Duck"               = "#CD853F",
  "Snow Goose"               = "#B0C4DE"
)

ggplot(data = monthly_waterfowl,
  mapping = aes(x = month,
    y = abundance,
    fill = species)) +

  # border lines between layers to reduce muddiness
  geom_area(color = "#1C2B2D", linewidth = 0.15, alpha = 0.92) +

  scale_fill_manual(values = waterfowl_colors) +

  scale_x_date(
    date_breaks = "1 year",
    date_labels = "%Y",
    expand = c(0, 0)
  ) +

  scale_y_continuous(expand = c(0, 0)) +
```

```

labs(
  title = "Monthly Waterfowl Abundance Through Time",
  x      = "Date",
  y      = "Monthly Abundance (individuals)",
  fill   = NULL,
  # note the large GWFG flock count for transparency
  caption = "Note: the January 2023 peak includes a single-point count of 91
Greater White-fronted Geese - a plausible but unusually large flock for this site."
) +

theme_minimal(base_size = 13) +
theme(
  plot.background = element_rect(fill = "#1C2B2D", color = NA),
  panel.background = element_rect(fill = "#243336", color = NA),
  panel.grid.major = element_line(color = "#3A4F52", linewidth = 0.4),
  panel.grid.minor = element_blank(),
  plot.title      = element_text(color = "#F0EAD6", size = 20, face = "bold",
                                margin = margin(b = 12), hjust = 0),
  plot.caption    = element_text(color = "#7A9A95", size = 8, hjust = 0,
                                margin = margin(t = 8)),
  axis.text       = element_text(color = "#B8C9C2", size = 10),
  axis.title      = element_text(color = "#D4C9A8", size = 11),
  axis.ticks      = element_line(color = "#3A4F52"),
  legend.position = "right",
  legend.background = element_rect(fill = "#1C2B2D", color = NA),
  legend.key       = element_rect(fill = NA, color = NA),
  legend.text      = element_text(color = "#F0EAD6", size = 8.5),
  legend.key.size  = unit(0.45, "cm"),
  legend.spacing.y = unit(0.1, "cm"),
  plot.margin      = margin(16, 20, 12, 16)
)

```

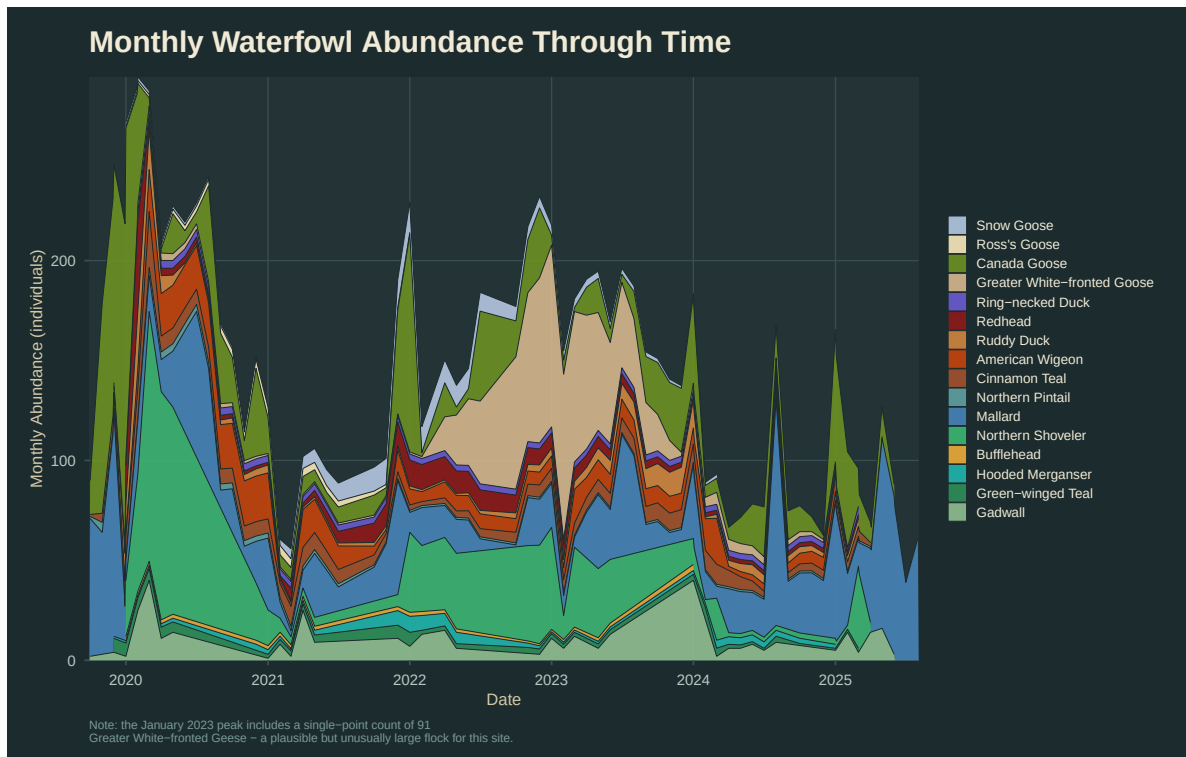


Figure 2 depicts total monthly waterfowl abundance at NCOS from water years 2020–2025, where each colored layer represents the summed monthly count of one waterfowl species, stacked to show total abundance across all species. Note: three species (Blue-winged Teal, Cackling Goose (Aleutian), Canvasback) are present in the data set but not visible due to very low counts (7 individuals total). Abundance is highest in early 2020 and generally declines through the study period, with peaks concentrated in winter months. This figure directly addresses our question of how waterfowl abundance varies through time at NCOS.

Figure 3: Lagged precipitation vs. total waterfowl abundance

```
ggplot(data = bird_weather_monthly,
       mapping = aes(x = lagged_prctp,
                     y = total_abundance,
                     # color by month of year (1-12) to show seasonal patterns
                     color = month_of_year)) +

  geom_point(alpha = 0.85, size = 4) +

  scale_color_viridis_c(
```

```

    name    = "Month",
    option  = "turbo",
    breaks  = c(1, 3, 5, 7, 9, 11),
    labels  = c("Jan", "Mar", "May", "Jul", "Sep", "Nov"),
    limits  = c(1, 12)
  ) +

  labs(
    title   = "Relationship Between Precipitation and Waterfowl Abundance",
    x       = "Previous Month's Precipitation (in)",
    y       = "Total Monthly Waterfowl Abundance (individuals)",
    caption = "Each point represents one month's waterfowl abundance plotted
against the previous month's precipitation. Points colored by month of year."
  ) +

  theme_minimal(base_size = 13) +
  theme(
    plot.background = element_rect(fill = "#1C2B2D", color = NA),
    panel.background = element_rect(fill = "#243336", color = NA),
    panel.grid.major = element_line(color = "#3A4F52", linewidth = 0.4),
    panel.grid.minor = element_blank(),
    plot.title      = element_text(color = "#F0EAD6", size = 20, face = "bold",
                                   margin = margin(b = 12), hjust = 0),
    plot.caption     = element_text(color = "#7A9A95", size = 8.5, hjust = 0,
                                   margin = margin(t = 8)),
    axis.text        = element_text(color = "#B8C9C2", size = 10),
    axis.title       = element_text(color = "#D4C9A8", size = 11),
    axis.ticks       = element_line(color = "#3A4F52"),
    legend.background = element_rect(fill = "#1C2B2D", color = NA),
    legend.key        = element_rect(fill = NA, color = NA),
    legend.text       = element_text(color = "#F0EAD6", size = 9),
    legend.title      = element_text(color = "#D4C9A8", size = 10, face = "bold"),
    legend.key.height = unit(1.2, "cm"),
    plot.margin       = margin(16, 20, 12, 16)
  )

```

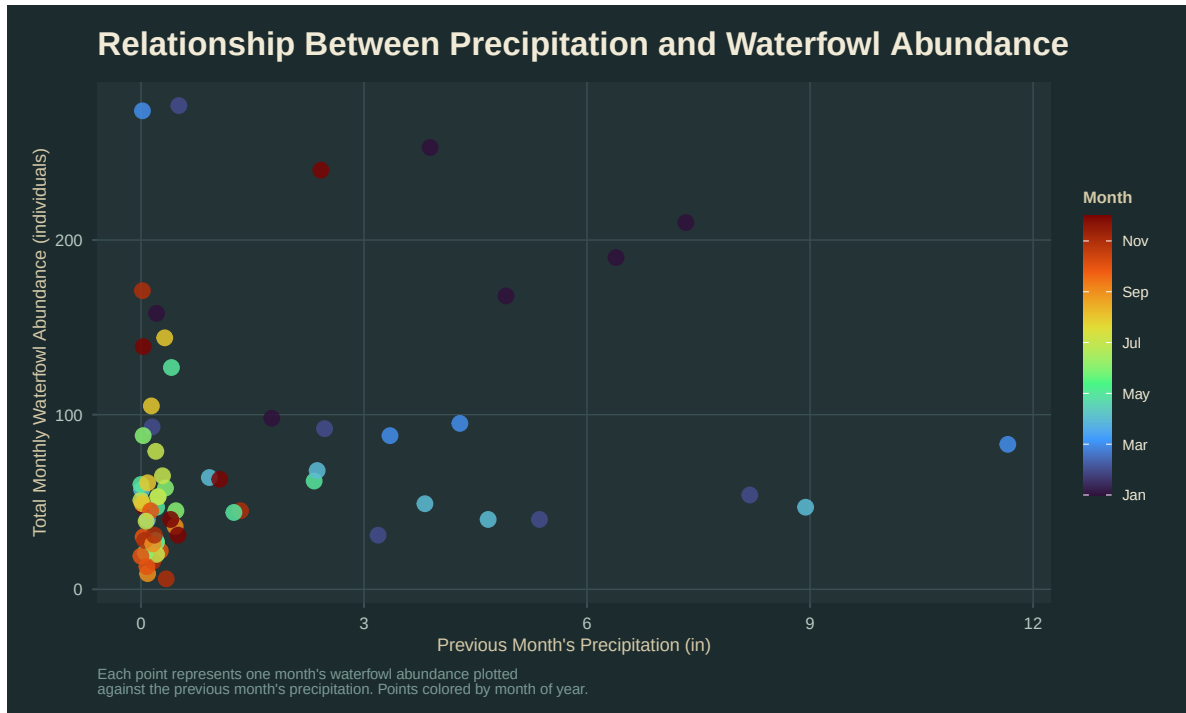


Figure 3 depicts the relationship between total monthly waterfowl abundance and the previous month's total precipitation at NCOS from water years 2020–2025, where each point represents one month colored by month of year. The figure shows no clear linear trend between lagged precipitation and waterfowl abundance, suggesting that the previous month's precipitation does not strongly predict waterfowl abundance at NCOS. This figure directly addresses our primary research question about the effect of precipitation on waterfowl abundance.

Other exploratory visualizations

Figure 4: Average seasonal trends comparing waterfowl abundance and precipitation

```
# shared theme
seasonal_theme <- theme_minimal(base_size = 13) +
  theme(
    plot.background = element_rect(fill = "#1C2B2D", color = NA),
    panel.background = element_rect(fill = "#243336", color = NA),
    panel.grid.major = element_line(color = "#3A4F52", linewidth = 0.4),
    panel.grid.minor = element_blank(),
    plot.title = element_text(color = "#F0EAD6", size = 14, face = "bold",
```

```

        margin = margin(b = 8), hjust = 0),
  axis.text   = element_text(color = "#B8C9C2", size = 10),
  axis.title  = element_text(color = "#D4C9A8", size = 11),
  axis.ticks  = element_line(color = "#3A4F52"),
  plot.margin = margin(12, 20, 8, 16)
)

# panel A
plot_abundance <- ggplot() +
  geom_point(
    data      = seasonal_by_year,
    mapping   = aes(x = season, y = seasonal_abundance, group = 1),
    color     = "#7EC8E3", alpha = 0.35, size = 2.5,
    position  = position_jitter(width = 0.08, seed = 42)
  ) +
  geom_line(
    data      = seasonal_summary,
    mapping   = aes(x = season, y = avg_abundance, group = 1),
    color     = "#7EC8E3", linewidth = 1.2
  ) +
  geom_point(
    data      = seasonal_summary,
    mapping   = aes(x = season, y = avg_abundance, group = 1),
    color     = "#7EC8E3", size = 4
  ) +
  labs(
    title = "Average Total Seasonal Waterfowl Abundance",
    x     = "Season",
    y     = "Abundance (individuals)"
  ) +
  seasonal_theme

# panel B
plot_precip <- ggplot() +
  geom_point(
    data      = seasonal_by_year,
    mapping   = aes(x = season, y = seasonal_prctp, group = 1),
    color     = "#6B8E23", alpha = 0.35, size = 2.5,
    position  = position_jitter(width = 0.08, seed = 42)
  ) +
  geom_line(
    data      = seasonal_summary,

```

```

    mapping = aes(x = season, y = avg_prctp, group = 1),
    color = "#6B8E23", linewidth = 1.2
) +
geom_point(
  data = seasonal_summary,
  mapping = aes(x = season, y = avg_prctp, group = 1),
  color = "#6B8E23", size = 4
) +
labs(
  title = "Average Total Seasonal Precipitation",
  x = "Season",
  y = "Precipitation (in)"
) +
seasonal_theme

# combine
combined_plot <- (plot_abundance / plot_precip) +
  plot_annotation(
    title = "Seasonal Patterns in Waterfowl Abundance and Precipitation",
    caption = "Points show individual water-year seasonal totals; lines show averages across",
    theme = theme(
      plot.background = element_rect(fill = "#1C2B2D", color = NA),
      plot.title = element_text(color = "#F0EAD6", size = 18, face = "bold",
                                hjust = 0, margin = margin(b = 10)),
      plot.caption = element_text(color = "#7A9A95", size = 8.5, hjust = 0,
                                  margin = margin(t = 8))
    )
  )
combined_plot

```

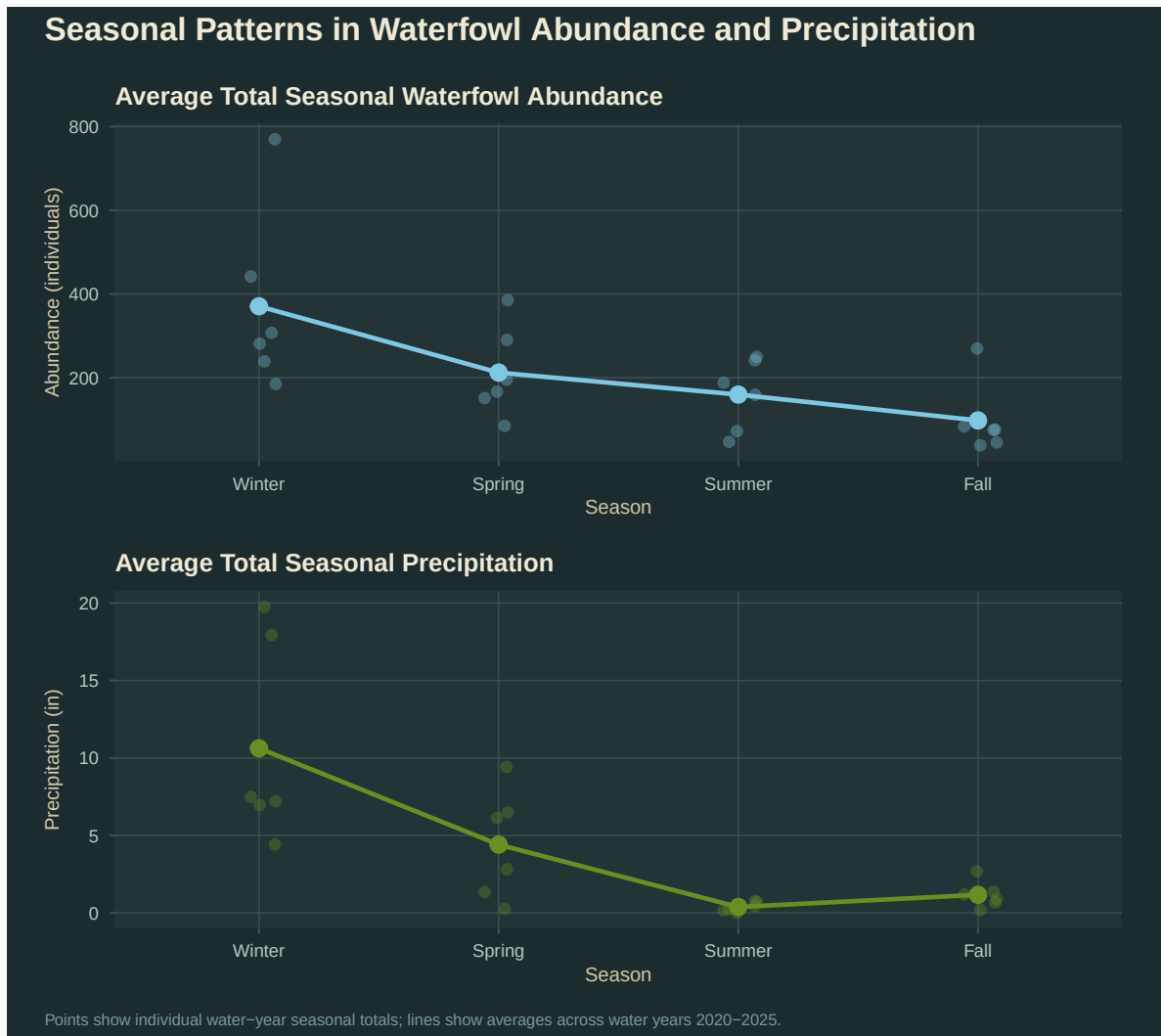



Figure 4 uses two stacked panels to compare average total seasonal waterfowl abundance and average total seasonal precipitation across water years 2020-2025. Small points show individual water-year seasonal totals to display the spread of observations; larger points and lines show the average across all water years. Both variables are highest in winter and lowest in summer, suggesting that waterfowl and precipitation follow similar seasonal patterns. However, Figure 3 indicates this parallel trend does not reflect a direct causal relationship. This figure contextualizes our main question by showing that seasonal timing, rather than precipitation amount, may better explain patterns in waterfowl abundance at NCOS.